Data pipelines, software, and source code

The massive difference between other approaches and the data mesh is that the team owning the data is also responsible for all of the *code* that created the data and its specific formats. So the team is responsible for the code of data pipelines, transformations, batch and stream processing, and migration scripts.

But source code is not enough to create working software. The team needs to follow DevOps culture and describe and spin up the required infrastructure and software delivery pipelines. The team members need to expose logging, monitoring, and tracking. In a nutshell, the team is responsible for the whole software delivery process, from understanding data the user needs, to code development and deployment, to feedback collection, to fixing production bugs and adding improvements.

Quality, cleansing, and deduplication

For specific purposes, data needs to meet a certain quality level. In the data mesh paradigm, the owning team is responsible for understanding consumer needs and providing the data that meets quality requirements. The data implementation team is responsible for cleansing and deduplication.

Enriching and aggregations

Sometimes users need a more aggregated form of information or information enriched with additional data. Some of these enrichments and aggregations are also in your scope.

Let’s look at the production cost statement example. A cost statement could be created weekly. However, suppose there is a requirement to prepare a yearly summary of all costs. In that case, the team owning that data product should create batch processing that periodically calculates all costs for a year and exposes it in the required form.

As you can see, with this approach, more responsibility is moved closer to the data-producing teams. In the following subsection, you’ll see how a similar shift in ownership and responsibility works at Messflix.